Teacher–Student Handover in Accelerated TTS Inference

A structured evidence review and mathematical synthesis of switching, blending, and bounded endpoint correction

Search cutoff: 19 September 2026

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Review designation: Structured evidence review with a prospective search protocol and targeted full-text screening; not an exhaustive systematic literature review
Scope: Inference formulas. Duration prediction and cadence calibration are held outside the proposed intervention.

> **Decision:** Retain endpoint relaxation as the principal research formulation. Compare a fixed acceptance weight with an adaptive displacement cap at the first student reconstruction. Treat cosine weighting as an ablation, not an established solution. The evidence does not justify declaring any candidate the empirically optimal "final formula."

Abstract

Background. The project initially framed teacher–student handover as an abrupt replacement of continuous velocity fields. Source inspection shows a more specific implementation: partial teacher integration, extrapolation to an endpoint estimate, and stochastic student renoising/reconstruction. The correct mathematical object therefore materially affects the proposed contribution.

Objective. To assess relevant formulas, distinguish evidence of numerical or distributional mismatch from evidence of audible artifacts, and select a defensible formulation for investigation under frozen-model constraints.

Methods. A protocol was recorded before external searches. Twenty-eight web-search queries, targeted primary-source retrieval, reference following, and limited forward searches were used. Seventeen technical works were retained for targeted full-text assessment, alongside PRISMA reporting guidance and three project sources. Extraction covered equations, assumptions, computational requirements, evidence directness, and transfer limitations. Ranked web results were not treated as exhaustive database exports; no meta-analysis or independent duplicate screening was performed.

Results. The literature supports investigating inference parameterization, input-distribution compatibility, and computation allocation. It does not establish that this project's handover causes audible noise. Continuous-field interpolation requires shared-state compatibility and can add network evaluations. Smooth weights do not establish distribution preservation or perceptual improvement. Existing relaxation theory supplies a mathematical precedent for partial updates, but its convergence assumptions are not verified for the stochastic student chain.

Recommendation. Investigate first-handover endpoint correction, with the adaptive candidate $E = T + \lambda(S - T)$, where $\lambda = \min\{1, \rho/\operatorname{RMS}(S - T)\}$ for a nonzero correction and $\lambda = 1$ otherwise. This is a projection onto a ball centered on the teacher endpoint, not a new general projection theorem. It adds no neural evaluations and guarantees a prescribed local displacement bound. Whether that restriction improves speech, preserves diversity, or merely suppresses useful student corrections must be tested.

Keywords: text-to-speech; flow matching; distillation; endpoint relaxation; teacher-guided sampling; inference; numerical analysis.

1. Introduction and bounded problem statement

The thesis needs a problem whose existence can be investigated and a formula whose consequences can be defended. An attractive interpolation curve is insufficient: the quantities being combined must mean the same thing, and the intervention must address a measured limitation.

The motivating task is to preserve useful teacher information while allowing an efficient student to refine a speech representation. Three distinct questions arise:

1. Is the teacher-to-student input compatible with the student's learned reconstruction behavior?
2. Is accepting the entire first student correction beneficial across inputs?
3. Can an inexpensive acceptance rule improve the quality–cost trade-off?

These questions allow a negative answer. A successful baseline may need no handover modification. A large correction may fix a bad teacher estimate. A smooth change in latent space may erase phonetic detail.

The **bounded problem statement** is therefore:

> In a frozen teacher-assisted TTS reconstruction chain, the relationship between the magnitude and direction of the first student endpoint correction and final synthesis quality has not been established by the inspected project evidence. This study should characterize that relationship and test whether bounded acceptance offers a useful quality–efficiency trade-off.

"Not established by the inspected evidence" is intentionally narrower than "never studied." This review cannot certify worldwide novelty.

2. Research questions

ID	Question	Main answer location
RQ1	What is actually switched or combined in the baseline?	Section 4
RQ2	What literature supports mismatch or error concerns?	Sections 5–6
RQ3	What properties follow from each candidate formula?	Sections 7–9
RQ4	What does "single computation" mean operationally?	Section 10
RQ5	Which formulation should the thesis investigate?	Sections 9 and 11
RQ6	What would establish a contribution or falsify the mechanism?	Sections 11–12

3. Review methodology

3.1 Protocol and scope

The protocol was recorded before external literature searching, after inspecting the current project specification and adviser report. The main date window was 2020 through 19 September 2026; earlier mathematical foundations were eligible. English-language accessible full texts were prioritized. Preprints were eligible but were distinguished from confirmed proceedings publications.

The review included: the exact baseline; flow and diffusion parameterizations; accelerated sampling; teacher/student reconstruction; relaxation; switched-system stability; and closely related adaptive guidance. It excluded unrelated uses of "teacher," "handover," and "consistency," secondary AI summaries as evidence, and mathematical conclusions available only from abstracts.

The extraction fields were fixed in advance: problem, original equation and location, normalized interpretation, assumptions, mathematical property, experimental support, computation, and transfer limitations. The synthesis was organized by mechanism rather than publication count.

3.2 Sources and search procedure

Discovery used a web search service, not authenticated Scopus, Web of Science, IEEE Xplore, or ACM Digital Library database exports. Primary texts were accessed through arXiv, ACL Anthology, PMLR, an author-hosted control paper, and official reporting guidance. Domain filters and title-specific searches were attempted; several queries returned substantial irrelevant material or publisher-root URLs. This limits sensitivity and reproducibility.

Appendix A contains the 28 exact submitted search strings. Search results led to full-text retrieval. The DMOSpeech 2 and DMD2 reference lists also directed attention to flow matching, consistency, and numerical-sampling work. Targeted forward searches used DMOSpeech 2, blending, handover, and trust-region terms. They are **not a complete forward-citation census**.

Versions were consolidated by title and arXiv identifier. Venue-search snippets were not accepted as definitive publication metadata when the actual proceedings record could not be resolved. The source version examined is distinguished from the possible later publication status.

3.3 Selection and accounting

Seventeen technical works were retained: fourteen generative-modeling/TTS works, two mathematical-foundation works, and one control-theory survey. PRISMA was a separate reporting reference. Three project sources were also inspected. Appendix B provides the retained-source and additional-candidate decisions.

These are **curated full-text-set counts**, not PRISMA identification or exclusion totals. All ranked search hits were not exported into a deduplicated bibliographic database. Accordingly, this manuscript does not invent the number of database records identified, duplicates removed, or title/abstract exclusions.

Full-text access means the document was obtained and relevant methods, equations, results, and limitations were assessed. It does not mean every appendix was independently reproduced. No unavailable full-text equation is used as a mathematical premise.

3.4 Appraisal and synthesis

Each work was assessed qualitatively for directness, mathematical transparency, empirical relevance, computation reporting, and applicability to frozen checkpoints. No pooled quality score was used: a correct theorem under unsuitable assumptions is not strong evidence for speech improvement.

No effect-size meta-analysis was appropriate because image FID, TTS WER, subjective ratings, diversity measures, and fixed-point residuals are different outcomes. Published performance numbers were not pooled or treated as comparable across hardware.

PRISMA 2020 is a reporting guideline with a 27-item checklist; using its principles does not make an incomplete search exhaustive. This review reports its limitations accordingly. PRISMA guidance

3.5 Deviations and limitations of the review process

There was one AI-assisted reviewer, no independent screening, no registered protocol, no librarian-reviewed search strategy, and no author contact. Adaptive searches added recent trust-region papers after that candidate emerged. Two recent candidates, dots.tts and the on-policy CFG study, were promoted from preliminary screening to full-text inclusion during the final coverage check. This is documented exploration, not a preregistered hypothesis test. The work is a substantive evidence review and analytical manuscript, but it must not be submitted as a completed exhaustive SLR without extending these procedures.

4. Baseline verification: what the project actually computes

4.1 Project evidence consulted

- **P1:** `DMOSpeech2_TAGS_Flow_Master_Project_Specification_Google_Docs.docx`, current version 2, particularly Sections 4, 10–12, and 17.
- **P2:** `TAGSFlow_Adviser_Progress_Report.md`, current version 1.
- **P3:** Uploaded `infer.py`, complete 559-line source snapshot. The functions `_teacher_sampling`, `_student_sampling`, and `generate` were inspected directly.

P1 already rejects the claim that the baseline simply changes velocity models along one uninterrupted ODE. P2 describes first-endpoint relaxation with a fixed student weight of 0.85. P3 verifies the underlying sampling operators, but does not contain that experimental assignment. The modified experimental implementation was not supplied in this review; no checkpoint or audio evaluation was executed.

4.2 Common notation

Symbol	Meaning
$x_t \in \mathbb{R}^d$	Noisy acoustic state at inference coordinate t
$t \in [0, 1]$	Noise-to-data coordinate: 0 is noise; 1 is data
c	Text, reference-audio conditioning, and applicable masks
v_T, v_S	Teacher/student network outputs in the flow parameterization used by P3
T	Cached teacher-produced endpoint estimate; not ground truth
S	First student-produced endpoint estimate
E	Endpoint accepted by the proposed rule
ϵ_k	Gaussian tensor drawn for student stage k
R_k	Stochastic student reconstruction operator
d	Number of valid coordinates in the compared normalized mel tensor

For this implementation, comparison occurs before multiplication by the checkpoint's mel scale. Tensors must have the same shape, frame order, conditioning, scale, and semantic interpretation. Equal shapes alone do not establish learned-distribution compatibility.

4.3 Source-derived teacher operator

For the deterministic teacher setting, P3 uses the guided output

$$v_T^{\mathrm{cfg}}(x, t, c) = v_T(x, t, c) + \gamma\big[v_T(x, t, c) - v_T(x, t, \varnothing)\big]. \tag{1}$$

The teacher integrates with Euler to its last retained grid point τ, then returns

$$T = x_\tau + (1 - \tau) v_T^{\text{cfg}}(x_\tau, \tau, c). \tag{2}$$

Equation (2) is a clean-endpoint **estimate**, not an exact integral to $t = 1$. With `eta > 0`, P3 adds a sampled perturbation inside its evaluated right-hand side; then the actual returned teacher quantity includes that perturbation. This code should not automatically be described as a standard SDE solver. Record the perturbations when reproducing it.

Grid detail found by source inspection and arithmetic: with `teacher_steps=16`, sway coefficient 1, and requested stop 0.07, the retained times are approximately

$$0,\ 0.00481527,\ 0.01921472,\ 0.04305966,\ 0.07612047.$$

The last retained time exceeds 0.07 because of the slicing rule. Four Euler updates plus the endpoint extrapolation require five right-hand-side evaluations. With separate conditional and unconditional teacher calls, that is ten teacher forward calls, followed by three student calls at the default start stage: thirteen calls in this static count. This is not a measured runtime or a claim that all published configurations use thirteen calls.

4.4 Source-derived student operator

For each retained student time t_k, P3 computes

$$z_k = (1 - t_k)\epsilon_k + t_k E_{k-1}, \qquad R_k(E_{k-1}; \epsilon_k, c) = z_k + (1 - t_k) v_S(z_k, t_k, c). \tag{3}$$

The baseline accepts $E_k = R_k(E_{k-1})$. For the first teacher-assisted stage, $E_0 = T$ and $S = R_1(T)$. Fresh noise is sampled at each stage.

Thus the baseline is an **operator composition**:

$$E_K = R_K \circ \cdots \circ R_1(T), \tag{4}$$

with conditioning and the realized random tensors understood. It is not established that v_S may be freely integrated at arbitrary off-grid states and times.

4.5 Paper versus implementation

The DMOSpeech 2 paper's Algorithm 2 expresses teacher state handoff followed by endpoint prediction and re-noising; Appendix B.4 likewise describes endpoint-style student generation. Its motivation is recovery of diversity lost under distillation, not a demonstrated audible discontinuity caused by handover. R1

P3 is more specific: it extrapolates a teacher endpoint and re-noises before the first retained student call. The paper's notation and the wrapper's operations are therefore not interchangeable. The published Algorithm 3 also draws an ϵ but writes z in its re-noising line; P3 unambiguously draws fresh noise. Reproduction should follow the chosen, versioned implementation and disclose the discrepancy rather than silently repair the paper.

4.6 What the prior preliminary results establish

P2 reports a 0.85 first-correction weight, conflicting peak-flux values, and different teacher endpoint norms before the intervention across compared runs. Those observations do not isolate a causal handover effect. A 15% reduction in the accepted displacement follows algebraically from that weight; improved naturalness does not. These internal results are motivation for replication, not an additional independent literature study.

5. Evidence map and source appraisal

Directness: D = exact baseline; A = adjacent TTS; G = other generative modeling; M = mathematical foundation. "Full text" below refers to relevant-section assessment, not independent verification of experimental claims.

ID	Work / inspected source	Directness	Useful evidence	Main transfer limitation	Appraisal
R1	DMOSpeech 2, arXiv 2025	D	Teacher guidance and student reconstruction	Paper/wrapper differences; no isolated switching-noise test	Direct mechanism evidence; causal gap
R2	F5-TTS, arXiv 2024	A	Flow representation and sway grid	Teacher sampling, not distilled handover	Strong schedule relevance
R3	DMD2, arXiv 2024	G	Multi-step input mismatch and training remedy	Images; remedy changes training	Strong adjacent mechanism
R4	Flow Matching, arXiv 2022	M/G	Conditional/marginal field construction	Does not justify arbitrary teacher/student mixtures	Strong conditional mathematics
R5	Rectified Flow, arXiv 2022	G	Regression and reflow for straighter transport	Requires learning; not a blending theorem	Relevant alternative mechanism
R6	DPM-Solver, 2022	G	Structure-aware integration	Diffusion parameterization conversion required	Strong numerical precedent
R7	Consistency Models, ICML 2023	G	Endpoint map and boundary conditions	Requires suitable training	Strong one-evaluation precedent
R8	CM-TTS, NAACL Findings 2024	A	Consistency parameterization in speech	Different trained system	Direct speech feasibility; weak plug-in transfer
R9	EPSS, arXiv 2025	A	Training-free grid pruning	No teacher/student endpoint blend	Important efficiency control
R10	MeanFlow, arXiv tech report 2025	G	Average-velocity identity	A new learned function, not weight merging	Strong alternative formulation
R11	Bravo et al., arXiv v2, 2017	M	Inexact relaxed fixed-point iteration	Nonexpansiveness/common-map assumptions unmet	Supports structure, not TTS convergence
R12	Liberzon–Morse, 1999 survey	M	Switching stability conditions	Equilibrium stability differs from speech quality	Foundational caution; not TTS evidence
R13	DP-DMD, arXiv 2026	G	First-step diversity supervision	Training intervention in images	Relevant recent counterproposal
R14	Delta Forcing, arXiv v1, 2026	G	Discrepancy-gated teacher supervision	Video chunks and training, not mel inference	Close conceptual prior art
R15	TRI-TSMC, arXiv v1, 2026	G	KL-constrained path-space adaptation	Particles and learned twisting functions	Distinguishes trust-region meanings
R16	dots.tts, technical report v1, 2026	A	CFG-aware interval-velocity distillation in speech	Different architecture and substantial training	Direct recent TTS alternative
R17	Rethinking CFG in OPD, preprint v1, 2026	G	Conditional-branch identification	Image/video training; not handover control	Conditioning compatibility caution

All seventeen documents were obtained. R12 was image-based; the relevant printed page 60 was visually inspected. No abstract-only record is used to support a central formula. Publication-status qualification and bibliographic links appear in Section 14 and Appendix B.

6. Formula-centered literature synthesis

The displayed published equations below retain source symbols where practical. Each following interpretation explicitly translates them to the project's noise-to-data convention. These are analytical comparisons, not proposed drop-in implementations.

6.1 F5-TTS: sampling coordinates are not acceptance weights

Published equation:

$$f_{\text{sway}}(u; s) = u + s\left[\cos\left(\frac{\pi u}{2}\right) - 1 + u\right]. \tag{5}$$

Location: F5-TTS, sampling discussion, Eq. (7). Here u is the original sampling coordinate and s controls its redistribution. This changes where a model is evaluated; it does not combine models. Speech experiments support the significance of evaluation placement. It can be applied without retraining, but transferring a new grid to a stage-trained student requires separate validation. R2

Project deduction: using a sway-generated time directly inside a cosine acceptance rule confounds two independent choices. At the first retained student time, $(1-\cos(\pi \times 0.07612047))/2 \approx 0.01422892$: only 1.42% of the correction is accepted. This arithmetic does not prove why an audio sample degraded.

6.2 DMD2: preserve the distinction between endpoint maps and velocities

Published updates:

$$\hat{x}_{t_i} = G_\theta(x_{t_i}, t_i), \qquad x_{t_{i+1}} = \alpha_{t_{i+1}}\hat{x}_{t_i} + \sigma_{t_{i+1}}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \tag{6}$$

Location: Sections 4.4–4.5. G_θ returns a denoised estimate; α, σ define the corruption schedule. The paper diagnoses training/inference input mismatch and uses generated training states to address it. It also distinguishes distribution matching from one-to-one trajectory imitation. This supports compatibility testing, not a claim that the student's output is the teacher's exact local tangent. Image evidence cannot establish TTS artifacts. Its training remedy is outside a frozen-checkpoint intervention. R3

Normalized interpretation: Eq. (3) is the project's reconstruction-and-corruption version of this operator pattern. Merely reusing a flow-shaped network output does not establish arbitrary-time ODE validity.

6.3 Flow matching: marginalization is a specific mixture, not arbitrary interpolation

Published equations:

$$u_t(x) = \int u_t(x \mid x_1)\frac{p_t(x \mid x_1)q(x_1)}{p_t(x)}\, dx_1, \tag{7}$$

$$\mathcal{L}_{\mathrm{CFM}} = \mathbb{E}_{t,q(x_1),p_t(x|x_1)}\|v_t(x) - u_t(x \mid x_1)\|^2. \tag{8}$$

Location: Eqs. (8)–(9), Theorems 1–2. The mixture weights depend on conditional densities; the objective learns the corresponding marginal field. These results support the shared ODE notation but do not establish that a time-only average of two independently learned fields preserves either endpoint distribution. R4

Project implication: no distribution-preservation theorem for cosine teacher/student weighting follows from conditional flow matching alone.

6.4 Rectified flow: straighten through learning

Published objective:

$$\min_v \int_0^1 \mathbb{E}\|(X_1 - X_0) - v(X_t, t)\|^2 dt, \qquad X_t = tX_1 + (1-t)X_0. \tag{9}$$

Location: Section 2.1, Eq. (1), inspected 2022 preprint. Here X_0, X_1 are endpoint samples. Reflow changes their coupling through repeated learning; the method is evaluated on image generation and transfer. It offers a path toward lower discretization difficulty, not a training-free guarantee that two existing fields form a straighter trajectory when averaged. R5

Normalized interpretation: this uses the same noise-to-data orientation as x_t. Adopting its learned transport mechanism would change the thesis's training budget.

6.5 DPM-Solver: derive from the operator being solved

Published exact representation:

$$x_t = \frac{\alpha_t}{\alpha_s} x_s - \alpha_t \int_{\lambda_s}^{\lambda_t} e^{-\lambda} \hat{\epsilon}_\theta(\hat{x}_\lambda, \lambda)\, d\lambda, \qquad \lambda_t = \log(\alpha_t/\sigma_t). \tag{10}$$

Location: Proposition 3.1, Eq. (3.4). Its original sampling direction has $t < s$; ϵ_θ predicts noise. Dedicated integration uses the semi-linear diffusion structure, with convergence-order analysis and image experiments. The practical precedent is structural derivation and explicit error assumptions. Its noise parameterization and schedule cannot be inserted unchanged into Eq. (3). R6

Normalized interpretation: reversing time is insufficient by itself; the noise prediction must also be mapped to a compatible flow field.

6.6 Consistency models and CM-TTS: one learned endpoint map

Published parameterization:

$$f_\theta(x,t) = c_{\mathrm{skip}}(t)x + c_{\mathrm{out}}(t)F_\theta(x,t), \quad c_{\mathrm{skip}}(\varepsilon) = 1, \quad c_{\mathrm{out}}(\varepsilon) = 0. \tag{11}$$

Location: Consistency Models, Section 3, Eq. (5). ε denotes the small terminal noise level, not a sampled Gaussian tensor. The learned map sends points on a probability-flow trajectory to a common endpoint; one-step generation depends on consistency training or distillation. This is an architectural boundary condition, not teacher/student averaging. R7

CM-TTS uses the same skip/output construction in its Eq. (5), with speech conditioning and consistency training. Its speech results demonstrate that such learned endpoint maps are relevant to TTS. They do not show that existing DMOSpeech 2 weights can become a consistency model by replacing their inference formula. R8

Normalized interpretation: define $D_S(z, t, c)$ as a learned clean-estimate map. One call to D_S is possible only if the checkpoint has learned the required mapping; notation cannot supply that capability.

6.7 EPSS: a relevant efficiency alternative

Published seven-evaluation grid:

$$\{t_i\} = f_{\text{sway}}\left(\{0, 1/16, 1/8, 3/16, 1/4, 1/2, 3/4, 1\}; -1\right). \tag{12}$$

Location: Section 3.2 and Table 2. Eight grid nodes produce seven Euler intervals. EPSS prunes the teacher's evaluation schedule and compares speech quality and timing. Its reported English WER moves from 1.70 to 1.74 while reducing RTF from 0.123 to 0.030 in its configuration: efficiency is a trade-off, not uniform improvement. It needs no retraining. It does not validate a distilled student's off-grid behavior. R9

Project implication: a new blend should face a schedule-only efficiency control where feasible; extra teacher computation must not be credited to the blend itself.

6.8 MeanFlow: average velocity is a different function

Published identities:

$$u(z_t, r, t) = \frac{1}{t-r}\int_r^t v(z_\tau, \tau)\, d\tau, \quad u = v - (t-r)\frac{d}{dt}u, \tag{13}$$

$$z_r = z_t - (t-r)u(z_t, r, t). \tag{14}$$

Location: Eqs. (3), (6), and (12). The source uses data at 0 and noise at 1. A network learns interval-average velocity, with the total derivative handled in training. One-evaluation generation is a learned capability, not the arithmetic mean of teacher and student velocities. Image experiments support that separate modeling approach. R10

Normalized interpretation: in forward noise-to-data time, an average field $\bar{v}(x_a, a, b)$ gives $x_b = x_a + (b-a)\bar{v}$. A teacher–student composite could potentially be distilled into such an interval map, but doing so requires training and a target construction.

6.9 Relaxation: a sound algebraic precedent with restricted convergence theory

Published inexact iteration:

$$x_{n+1} = (1-\alpha_{n+1})x_n + \alpha_{n+1}(Tx_n + e_{n+1}). \tag{15}$$

Location: Bravo et al., Section 1, equation labeled `(ikm)`; Theorem 1.1. The map T is nonexpansive, with further boundedness and error conditions for stated results. Here T is an operator, unlike our teacher endpoint symbol. Mapping this structure to $E_k = E_{k-1} + \lambda_k(R_k(E_{k-1}) - E_{k-1})$ provides a relaxation interpretation. Stage-varying stochastic neural operators do not automatically satisfy the theorem. R11

6.10 Switching theory: smoothness and stability are different

Published system family: $\dot{x} = f_p(x),\ p \in \mathcal{P}$, Eq. (3), printed p. 60. The discussion gives a common Lyapunov-function condition and illustrates that stable subsystems can produce unstable switching. This supports checking assumptions, not predicting speech artifacts. Equilibrium stability is also different from accuracy over a finite generative trajectory. The article is a foundational survey, not a new empirical TTS study. R12

6.11 Recent alternatives constrain novelty claims

DP-DMD. The inspected paper constructs a teacher-derived first-step target

$$\tilde{v} = \frac{\epsilon - z_{\tilde{t}}}{1 - \tilde{t}}, \qquad \ell_{\text{Div}} = \mathbb{E}_\epsilon \| v_\theta(\epsilon, 1) - \tilde{v} \|^2. \tag{16}$$

Location: Eqs. (6)–(7), Section 4.2. Noise is at 1. First-step supervision is separated from later distribution matching through stopped gradients. It is image-domain training evidence that early-stage preservation is already an active design direction; it is not an inference-only endpoint cap. R13

Delta Forcing. In the inspected v1, Eq. (11) defines

$$\rho_k = \| \delta_k^{\text{real}} - \delta_k^{\text{fake}} \|_2, \qquad w_k = \text{sigmoid}[-s(\rho_k - \mu)]. \tag{17}$$

The feature deltas concern video evolution; μ, s control gating. It modulates teacher supervision during training and reports video ablations. This is conceptually close prior art for adaptive reliance on a teacher, but its features, time axis, objectives, and costs differ from a TTS endpoint correction. R14

TRI-TSMC. Eqs. (4.5)–(4.6) give

$$q_i^* = \arg\min_q D_{KL}(q \| \pi) \quad \text{s.t. } D_{KL}(q \| P_i) \le \varepsilon, \qquad \frac{dq_i^*}{dP_i} \propto \left(\frac{d\pi}{dP_i} \right)^{1/(1+\lambda_i)}. \tag{18}$$

These are path distributions, not mel tensors. The method uses particles and fits twisting functions, although base-model weights remain fixed. Its text/image results cannot justify calling our tensor projection a KL trust region or a fully equivalent algorithm. R15

6.12 Recent TTS interval distillation and guidance compatibility

dots.tts. The technical report's Eqs. (13)–(14) construct a guided teacher interval target and fit a student:

$$\bar{v}^{T,\text{cfg}}_{t_a \to t_b} \approx \frac{x^{T,\text{cfg}}_{t_b} - x^{T,\text{cfg}}_{t_a}}{t_b - t_a}, \qquad \ell_{\text{mv}} = \| v_\phi(x_{t_a}, t_a, \Delta t, c) - \bar{v}^{T,\text{cfg}}_{t_a \to t_b} \|_2^2. \tag{18a}$$

Its Section 3.2.4 updates $x_{t_b} = x_{t_a} + \Delta t\, v_\phi$ with one conditional call per interval. The reported system evaluates 2–4 calls, not a demonstrated one-call complete utterance. Training adds interval conditioning and uses teacher trajectories. This is a concrete speech-domain path to absorbing teacher guidance into student inference, but not an arithmetic retrofit of the current checkpoints. R16

Guidance compatibility. Li et al.'s Eq. (10) instead supervises positive prediction and conditional direction separately:

$$\ell_{\mathrm{PDM}} = \|e_+\|_2^2 + \lambda\|e_+ - e_-\|_2^2, \qquad \lambda > 0, \tag{18b}$$

where e_+, e_- are teacher-minus-student positive/negative branch errors. Its image/video experiments distinguish benign shared conditioning from harmful privileged negative conditioning. This supports auditing guidance conventions; it does not diagnose the same failure in DMOSpeech 2. The project teacher has explicit CFG branches whereas its student wrapper has a single conditional call, so "same conditioning" must be checked semantically, not only by argument names. R17

7. What the evidence supports—and what it does not

Proposition	Assessment	Implication for the thesis
Evaluation placement and parameterization can affect generative accuracy	Supported in adjacent work	Record actual grids and output semantics
A distilled model can be sensitive to training/inference input mismatch	Supported; strongest adjacent mechanism is R3	Probe actual reconstruction inputs
Teacher guidance is intended to retain useful diversity	Direct baseline motivation, R1	Include diversity among outcomes
The current handover causes audible noise	Not established by retained evidence	Do not use this as an asserted gap
Cosine weighting guarantees better speech	Unsupported	Test against fixed/linear/polynomial rules
A smaller endpoint correction is necessarily more accurate	False in general	Include counterexamples and rejection criteria
One equation implies one network evaluation	False	Count actual forwards and guidance branches
Known interpolation alone establishes novelty	Unsupported	Contribution needs characterization, justified control, and validation

A velocity jump changes a derivative; it need not jump the state. A re-noising operation deliberately changes the state distribution; it is not automatically an integration defect. Numerical error is discrepancy from a specified differential-equation solution. Acoustic distortion concerns the decoded signal. Audible noise is a perceptual outcome. None of these implications should be assumed without an intervening argument or experiment.

The defensible literature foundation is consequently **operator compatibility and bounded intervention**, not a categorical assertion that abrupt switching is harmful.

8. Critical mathematical comparison

The propositions and examples in this section are derivations for this review, not claims that a cited paper tested the proposed TTS method.

8.1 Hypothetical hard switch and continuous blend

For compatible fields at the same state, a hard switch is

$$v_H(x,t) = \begin{cases} v_T(x,t), & t < \tau, \\ v_S(x,t), & t \geq \tau. \end{cases} \tag{19}$$

With a continuous state and no reset, a solution can remain continuous while its derivative changes at τ. By contrast, the project uses the resets/reconstructions of Eq. (3); Eq. (19) is a separate experimental model.

For $a < b$, define $q = \mathrm{clip}((t-a)/(b-a), 0, 1)$ and

$$v_B = (1 - w(q))v_T + w(q)v_S. \tag{20}$$

This requires matching state, time convention, scale, conditioning, and prediction type. It requires both models at every genuinely mixed evaluation unless one output is already available at exactly that input.

8.2 Weight families and endpoint regularity

Weight inside $0 < q < 1$, extended by constants outside	Endpoint values	Endpoint slopes	Regularity of extended weight	What it does not guarantee
Hard step	$0 \to 1$	Undefined at switch	Discontinuous	Accuracy or stability
Linear q	0, 1	1, 1 inside	C^0, not generally C^1	Smooth velocity derivative
Cosine $(1 - \cos \pi q)/2$	0, 1	0, 0	C^1, not generally C^2	Best error or audio quality
Cubic $3q^2 - 2q^3$	0, 1	0, 0	C^1, not generally C^2	Advantage over cosine
Quintic $6q^5 - 15q^4 + 10q^3$	0, 1	0, 0; second derivatives also 0	C^2	Benefit at sparse stages
Fixed endpoint weight λ	Not a time transition	Not applicable	An algebraic acceptance rule	Continuous-trajectory claims
Residual cap from Section 9	Data-dependent	Kink at cap boundary	Continuous projection in S for fixed T	Perceptual reliability

These polynomial and trigonometric properties follow by direct differentiation. They are standard interpolation constructions; no discovery claim is made.

At fixed x,

$$\partial_t v_B = (1-w)\partial_t v_T + w\partial_t v_S + w'(t)(v_S - v_T). \tag{21}$$

If both fields are jointly C^1, the weight has matching endpoint values and zero boundary slope, and all definitions agree at the boundaries, the extended blended field is C^1. If the fields are merely continuous and locally Lipschitz in state, weaker solution regularity is available. Neither statement automatically covers the stochastic chain or its decoded waveform.

For cosine,

$$\max_t |w'(t)| = \frac{\pi}{2(b-a)}. \tag{22}$$

A narrow transition can therefore create a large interior derivative proportional to model disagreement. Smooth endpoints do not eliminate a numerical resolution requirement.

For a sufficiently smooth deterministic field, the leading Euler local truncation term is

$$\frac{h^2}{2}\left(\partial_t v_B + D_x v_B\, v_B\right) + O(h^3). \tag{23}$$

A theorem comparing audio or global numerical error does not follow from eliminating a derivative jump alone.

8.3 Smooth convex blending need not preserve stability

Consider two illustrative linear fields with matrices

$$A_T = \begin{pmatrix} -1 & 4 \\ 0 & -1 \end{pmatrix}, \qquad A_S = \begin{pmatrix} -1 & 0 \\ 4 & -1 \end{pmatrix}.$$

Each has eigenvalues 1. Their constant 50–50 blend has matrix $\begin{pmatrix} -1 & 2 \\ 2 & -1 \end{pmatrix}$, with eigenvalues 1 and 3. The blended field is smooth but unstable. This is a mathematical counterexample to a general guarantee, not a model of measured TTS behavior. A common Lyapunov argument would require additional verified conditions.

8.4 Coarse-stage identifiability

Under P3, the default retained student times are approximately $0.07612, 0.29289, 0.61732$. A transition interval with no interior evaluated point has no resolved gradual blend. If only the first endpoint is modified, any time schedule is observed at one point and is indistinguishable from its resulting scalar weight.

Even with three stages, index-based progress $q = 0, 0.5, 1$ gives identical values for linear, cosine, cubic, and quintic schedules: 0, 0.5, 1. Such an experiment cannot distinguish their shapes. Normalizing the actual retained times instead gives cosine weights approximately $0, 0.34630, 1$. The coordinate must be specified before claiming a scheduling effect.

A floor-weight cosine,

$$w(q) = w_{\min} + (1 - w_{\min})\frac{1 - \cos(\pi q)}{2}, \tag{24}$$

avoids a near-zero first acceptance. But $w(0) = w_{\min} > 0$. If it is attached to a pure teacher field before the interval, it generally creates a boundary mismatch unless the fields agree there. It cannot simultaneously be advertised as a teacher-only-to-student-only C^1 transition without another construction. As an endpoint rule, no such continuous-field claim is needed.

8.5 Shared-state endpoint and velocity blending: exact scope of equivalence

Define clean estimates at **the same** noisy state:

$$D_T(z,t) = z + (1-t)v_T(z,t), \qquad D_S(z,t) = z + (1-t)v_S(z,t).$$

Then

$$(1-w)D_T + wD_S = z + (1-t)[(1-w)v_T + wv_S]. \tag{25}$$

This algebraic equivalence is useful only when z, t, conditioning, and scaling agree. A cached teacher endpoint from another state and a student endpoint after fresh re-noising do not satisfy those premises. Their interpolation is endpoint relaxation, not proof of velocity continuity.

8.6 Candidate comparison under frozen-model constraints

Candidate	Intervention	Extra neural work relative to P3	Principal risk	Role
Baseline full acceptance	$E = S$	None	Any existing weakness remains	Mandatory control
Fixed first relaxation	$E = T + \lambda(S - T)$	None	Withholds useful correction	Primary simple baseline
Scheduled relaxation	Moving anchor E_{k-1}, chosen λ_k	None if cached endpoints used	Confounds stages; indistinguishable schedules	Secondary ablation
Shared-state velocity blend	Eq. (20)	Teacher and student calls at mixed states	Unsupported student inputs; cost	Conditional experiment
Repeated cached-teacher anchor	Blend each S_k back toward original T	None	Repeated anchoring changes the intended chain	Separate method; do not conflate
Bounded first correction	Section 9	Norm and vector arithmetic only	Correction magnitude is not error	Recommended research candidate
Learned gate	Predict reliability from features	Feature/gate cost; possible training	Calibration and overfitting	Later extension
Consistency/MeanFlow/re-distillation	Learn an endpoint or interval map	Substantial training	Outside current frozen-weight scope	If one total network call is required

9. Recommended formulation and derivation

9.1 Recommendation and degree of commitment

The evidence justifies choosing an **operator-level research family** more strongly than a unique weighting function. Begin with the first-handover relaxation already suggested by P2. Compare a fixed weight against an adaptive cap. Do not make cosine the thesis's required centerpiece before this comparison.

The adaptive formulation below is selected for its interpretability, alignment with P3, exact geometric property, and absence of extra neural evaluations. Its selection is a reasoned design recommendation, **not a literature-established ranking of synthesis quality**.

9.2 Complete proposed operator

Compute the baseline teacher endpoint T and first student proposal

$$z = (1 - t_1)\epsilon_1 + t_1 T, \qquad S = z + (1 - t_1)v_S(z, t_1, c), \qquad D = S - T. \tag{26}$$

For one utterance, flatten the valid, unpadded normalized mel coordinates being updated. Let

$$r = \mathrm{RMS}(D) = \frac{\|D\|_2}{\sqrt{d}}, \qquad \rho \geq 0. \tag{27}$$

Define

$$\lambda_\rho(D) = \begin{cases} 1, & r = 0, \\ \min(1, \rho/r), & r > 0, \end{cases} \qquad \boxed{E = T + \lambda_\rho(D)(S - T).} \tag{28}$$

Apply Eq. (28) **once**, at the first teacher-assisted student reconstruction. Accept subsequent student reconstructions exactly as in the baseline, with the same stage grid. The first implementation should retain the baseline conditioning and prompt handling; changing masks or clamping prompt coordinates is a separate intervention.

ρ has units of normalized mel-coordinate displacement per coordinate. It is not an acoustic-noise threshold. RMS reduces the mechanical dependence of a raw L2 norm on tensor length, but does not guarantee comparability across speakers, normalization schemes, or different compositions of prompt and target frames. For batches, compute it separately for each utterance with an explicit valid-coordinate mask.

Boundary cases: $\rho = 0$ accepts no nonzero correction; $r \leq \rho$ exactly recovers baseline acceptance; $D = 0$ returns $T = S$; a sufficiently large radius recovers the baseline for all finite proposals.

9.3 Derivation as constrained closest-proposal selection

Let the allowed endpoint set be

$$\mathcal{B}(T, \rho) = \{Y : \|Y - T\|_2 \leq \rho\sqrt{d}\}.$$

Choose the feasible endpoint closest to the student's proposal:

$$E = \arg\min_{Y \in \mathcal{B}(T,\rho)} \frac{1}{2}\|Y - S\|_2^2. \tag{29}$$

If S lies inside the ball, $E = S$. Otherwise, the closest point lies on the radial segment from T to S, at distance $\rho\sqrt{d}$ from T. Substituting its radial coefficient gives Eq. (28). This is ordinary Euclidean projection, not a novel general formula or a perceptual optimization objective.

The literature contributions are distinct: P3 supplies the reconstruction operator; R11 supplies a relaxation precedent; R3 motivates compatibility checks; R14–R15 show that constrained/adaptive teacher influence already has related uses. **None of these sources supplies or validates this exact TTS radius choice.** The radius-constrained design is the review's proposed adaptation, with its derivation shown here.

9.4 Properties that can be proved

Proposition 1: local displacement bound.

$$\mathrm{RMS}(E - T) = \min\{r, \rho\}. \tag{30}$$

Proof: $E - T = \lambda D$, with $0 \le \lambda \le 1$. Substituting the two cases in Eq. (28) proves the result.

Proposition 2: baseline-deviation identity.

$$\mathrm{RMS}(E - S) = \max\{0, r - \rho\}. \tag{31}$$

This measures how much of the student proposal is withheld, not how much error is removed.

Proposition 3: nonexpansiveness in the proposal with a fixed teacher endpoint. Euclidean projection onto a fixed closed convex ball satisfies

$$\|P_{\mathcal{B}}(S_1) - P_{\mathcal{B}}(S_2)\|_2 \le \|S_1 - S_2\|_2. \tag{32}$$

One derivation uses the two projection optimality inequalities; adding them gives $\|E_1 - E_2\|^2 \le \langle E_1 - E_2, S_1 - S_2\rangle$, then Cauchy–Schwarz gives Eq. (32). This does not assert nonexpansiveness of the full pipeline when T, noise, or conditioning changes.

Proposition 4: conditional propagation bound. Couple all later noise draws and assume each later reconstruction map has a valid local Lipschitz bound L_k over the intervening states. If $F = R_K \circ \cdots \circ R_2$, then

$$\|F(E) - F(S)\|_2 \le \left(\prod_{k=2}^{K} L_k\right) \sqrt{d}\, \max\{0, r - \rho\}. \tag{33}$$

This is a sensitivity bound, not a quality guarantee. The constants have not been established for the checkpoint. For fixed noise and differentiable v_S, Eq. (3) gives

$$D_E R_k = t_k I + t_k(1 - t_k) D_z v_S, \tag{34}$$

which suggests local Jacobian-vector probes. A small number of such probes is a diagnostic, not a global Lipschitz certificate.

9.5 Why smaller corrections can be worse

For an illustrative aligned target Y, let $e_T = T - Y$. Then

$$\|E - Y\|^2 = \|e_T\|^2 + 2\lambda\langle e_T, D\rangle + \lambda^2\|D\|^2. \tag{35}$$

The best segment weight for this squared-error target, when $D \neq 0$, is

$$\lambda^* = \text{clip}\left(-\frac{\langle e_T, D\rangle}{\|D\|^2}, 0, 1\right). \tag{36}$$

Correction magnitude alone does not reveal the required inner product. If the student is already correct, shrinking its correction hurts. Moreover, zero-shot speech admits multiple valid realizations, so a single reference mel is not a universal perceptual target. Equation (36) is an explanatory oracle, not an inference rule.

The residual also includes re-noising effects:

$$D = (1 - t_1)\left[\epsilon_1 + v_S((1 - t_1)\epsilon_1 + t_1 T, t_1, c) - T\right]. \tag{37}$$

Consequently, $\|D\|$ must not be labeled "teacher–student velocity error" or "noise severity." It is a reconstruction displacement containing stochastic and model-dependent contributions.

9.6 Worked example: illustrative, not experimental

Let $T = (2, -1)$, $S = (5, 3)$, so $D = (3, 4)$, $d = 2$, and $r = 5/\sqrt{2} \approx 3.53553$. Set $\rho = 2$.

Quantity	Calculation	Result
Acceptance	$\lambda = 2/(5/\sqrt{2})$	0.565685
Accepted endpoint	$(2, -1) + 0.565685(3, 4)$	(3.697056, 1.262742)
Accepted RMS displacement	λr	2.000000
Withheld RMS displacement	$(1 - \lambda)r$	1.535534
Additional neural evaluations	Arithmetic after T, S exist	0

If $Y = S$, the baseline squared error is zero, while the capped endpoint has squared error $(5 - 2\sqrt{2})^2 \approx 4.71573$. The cap's exact geometric guarantee coexists with worse target error in this example.

9.7 Implementation pseudocode

```
# Proposed first-handover modification; not a validated speech implementation.
# T, S: matching normalized mel tensors for one utterance.
# valid: mask for valid coordinates; retain baseline prompt handling.
# eps1 and later noises are pre-generated and shared across paired runs.
T = baseline_teacher_endpoint(inputs, teacher_noise)
z = (1 - t1) * eps1 + t1 * T
S = z + (1 - t1) * student(z, t1, conditioning)
D = S - T
r = sqrt(mean(square(D[valid])))
lam = 1.0 if r == 0.0 else min(1.0, rho / r)
E = T + lam * D
for t, eps in remaining_baseline_stages_and_noises:
    z = (1 - t) * eps + t * E
    E = z + (1 - t) * student(z, t, conditioning)
return baseline_decode(E)
```

Check tensor finiteness and valid-coordinate count before computing the radius. Such checks catch invalid arithmetic; they do not certify audio quality. Include the bypass $E = S$ and fixed λ modes in the same instrumentation path.

9.8 Failure modes and rejection conditions

Reject or narrow the method if the teacher endpoint is poor, the student needs a large corrective update, norm magnitude does not identify harmful cases, or the cap systematically reduces intelligibility or diversity. Further risks include mixing semantically misaligned endpoints, scale drift, padding contamination, normalization-dependent thresholds, and downstream amplification. The projection itself has a derivative kink at its activation boundary; the recommendation prioritizes an exact displacement limit, not a C^1 claim.

10. What "single computation" can honestly mean

Interpretation	Feasible here?	Required work
One expression containing teacher and student contributions	Yes: Eq. (28), with Eq. (26) substituted	Teacher endpoint and student proposal still computed
One endpoint-acceptance operation	Yes	Vector arithmetic after both quantities exist
One student forward call after a cached teacher endpoint	Yes for the first reconstruction	Teacher cost already paid; later student stages remain
One total neural call from initial noise to final acoustic output	Not established for this frozen pipeline	Suitable trained one-step map or distillation
One teacher-plus-student solver evaluation	Can be expressed as one composite function	Usually at least two model evaluations; CFG may add more
Student-only inference incorporating learned teacher behavior	Possible in principle	Existing or newly trained compatible student; validate independently

The full proposed first-acceptance expression is

$$E = T + \lambda_\rho(D)\underbrace{[(1-t_1)\epsilon_1 + t_1 T + (1-t_1)v_S((1-t_1)\epsilon_1 + t_1 T, t_1, c) - T]}_{D}, \tag{38}$$

where λ_ρ is computed from the same bracketed residual. This encapsulates a cached teacher result and a student correction in one operator. It does not fuse model weights or eliminate the computation that produced T.

For nonlinear networks, averaging parameters generally does not equal averaging outputs. An ensemble packaged inside a single software call still performs both networks' work. If one total evaluation is an adviser requirement, the team should explicitly reconsider the frozen-weight constraint and pursue a learned endpoint/average-velocity map rather than promise arithmetic fusion.

11. Proposed validation plan

11.1 Gate 0: establish an executable and traceable baseline

Recover checkpoints, training configuration, exact source revision, tokenizer, normalization scale, vocoder, dependency versions, and evaluation data. Verify evaluation mode and log any repairs separately from the proposed method. P3's teacher-only wrapper has interface issues, and the

stopping-time slicing deserves explicit tests at boundary values. A repaired baseline must be applied consistently to all relevant conditions.

Pre-generate the teacher initial noise, teacher perturbation sequence, and every student noise tensor. Same seed alone is insufficient if different code paths consume random numbers differently. Fix text, reference audio, duration, and conditioning across each paired comparison. Keep the duration predictor outside the intervention.

11.2 Gate 1: characterize the problem before tuning a remedy

For each item, log T, S, D, r, accepted weight, actual times, teacher/student forwards, failure flags, waveform output, and final quality measures. Examine whether large r predicts a higher rate of failures or poorer final outputs after controlling descriptively for duration and speaker. This is hypothesis generation, not proof that r measures error.

For a more direct mechanism test, run several acceptance weights on the **same** T, S and all the same later noise tensors. Determine whether withholding correction actually helps those items. The acceptance intervention, not residual correlation alone, tests the proposed causal mechanism.

11.3 Baselines and ablations

ID	Condition	Purpose
B0	Student only	Efficiency/diversity context
B1	Original teacher-guided full acceptance	Primary matched comparison
B2	Reproduced teacher only	Quality/cost reference
A1	First weight $\lambda \in \{0.70, 0.85, 0.95, 1\}$ on development data	Does simple relaxation suffice?
A2	First adaptive cap, with development-selected ρ and baseline bypass	Does data-dependent acceptance help?
A3	Linear/cosine/cubic/quintic schedules on an explicitly specified coordinate	Does schedule shape add value?
A4	First-only versus moving-anchor multi-stage relaxation	Does intervention location matter?
A5	Shared-state field blend, only after compatibility checks	Alternative mechanism; count added calls
A6	Teacher schedule-only control, including feasible EPSS-style settings	Is benefit attributable to computation placement?

Do not describe A3 as a shape comparison if all sampled weights coincide. Do not compare a repeatedly reused teacher anchor against a moving-anchor method without identifying the change.

For ρ, use a small prespecified set of development residual quantiles plus the no-cap condition; lock the chosen value before final evaluation. A quantile is a tuning device, not an estimate of acceptable error. Use equal tuning budgets for the fixed and adaptive rules.

11.4 Feasible data design

A resource-conscious starting design is 30 development prompt–text items and 90 held-out items, spanning at least ten speakers and short, medium, and long texts, with three coupled seeds per

item. These are proposed planning quantities, not a power calculation or a promise of adequate statistical precision. Use checkpoint-supported languages and separate speakers/texts between tuning and final evaluation where feasible.

Run a small pilot to estimate variance and time cost. Finalize sample size from the smallest effect the team considers meaningful and the obtainable confidence interval; document any resource limitation. Avoid choosing the sample size by repeatedly checking significance.

11.5 Outcomes and claims

Claim being tested	Measurement	Necessary interpretation
Improved perceptual quality	Blinded paired preference or naturalness ratings	Primary human endpoint; automated predictors are supplementary
Preserved intelligibility	Fixed ASR and WER with fixed text normalization	ASR errors are an imperfect proxy; inspect catastrophic failures
Preserved speaker identity	Fixed speaker-embedding similarity	Report model/version; not equivalent to listener identity judgment
Preserved diversity	Across-seed pitch/prosody and embedding variation	Hold duration fixed; neither more nor less variance is automatically better
Controlled first displacement	r, λ, accepted/withheld RMS	Verifies the operator property, not quality
Reduced downstream sensitivity	Coupled-noise endpoint perturbation tests	Local behavior; no global stability theorem
Efficient inference	Teacher/student calls, wall time, RTF, peak memory	Warm-up, synchronized timing, batch/precision/hardware recorded

Spectral flux may be an exploratory diagnostic, but minimizing it can flatten legitimate speech transitions. Denoising time is not waveform playback time: a change at one inference stage affects the evolving whole spectrogram. It should not be presented as an audible splice at a corresponding audio timestamp.

11.6 Statistical analysis and decision rule

Choose one primary final comparison: adaptive cap versus the strongest fixed-weight development winner, with B1 retained as a baseline. Human evaluation should randomize order and conceal method labels. Treat utterances and speakers as clustered observations; multiple seeds and ratings are not automatically independent samples. Use paired estimates and a hierarchical or cluster bootstrap, with uncertainty reported for quality differences and timing ratios. Correct exploratory multiple comparisons or clearly label them exploratory.

Before final evaluation, specify intelligibility and speaker-similarity non-inferiority margins in meaningful units; select them with adviser input and pilot variability rather than inventing universal tolerances. Likewise specify the minimum worthwhile perceptual effect and acceptable runtime overhead.

Retain the adaptive method only if it improves the prespecified primary outcome relative to B1 and offers a defensible advantage over fixed relaxation, while satisfying secondary constraints. If it only bounds displacement, report exactly that. If the fixed weight performs as well, prefer the simpler fixed method. If neither helps, retain full acceptance and revise the claimed mechanism.

12. Theoretical contribution and novelty assessment

The following are already known mathematical ingredients: convex interpolation, cosine easing, polynomial boundary matching, relaxed iteration, and projection onto a Euclidean ball. Their appearance in an equation is not sufficient novelty.

A defensible contribution could combine:

1. **Correct operator characterization:** distinguish teacher endpoint extrapolation from stochastic student reconstruction, with reproducible instrumentation.
2. **Problem evidence:** establish when the first correction is beneficial or harmful under controlled interventions.
3. **A justified acceptance policy:** select a computable rule from diagnostics, derive its local property, and delimit its assumptions.
4. **Cost-aware validation:** demonstrate a reproducible quality benefit at unchanged neural-evaluation count and report actual overhead.
5. **A useful negative finding:** show that smooth schedules are unidentifiable on the chosen grid, or that correction magnitude is an inadequate reliability signal.

For a theoretical thesis, a local projection bound alone is elementary. A stronger analytical contribution would require an original, relevant condition relating reconstruction mismatch to downstream error, a justified data-dependent bound, or a consistency/error result for the actual stochastic operators. Such a result must not assume the desired conclusion or substitute a teacher endpoint for ground truth without justification.

Provisional contribution statement:

> This study characterizes teacher-assisted reconstruction in distilled TTS and evaluates bounded acceptance of the first student endpoint correction. It derives local displacement and conditional sensitivity properties and tests whether these controls improve perceptual quality at unchanged neural-evaluation count.

This statement is a research objective until experiments are complete. The term "trajectory-aware" should be retained only if the method actually uses trajectory information; a single residual norm is more accurately described as displacement-aware or residual-bounded. No final title is required to commit to cosine.

13. Threats to validity and conclusion

The search was limited to accessible web discovery and selected primary full texts. It was not exhaustive, and later or differently indexed prior art may change the novelty assessment. Several works were inspected as preprints; their equations and reported results were not independently reproduced. The review's own derivations are conditional and have not undergone external mathematical review.

The project source snapshot may differ from the checkpoint's original training and published implementation. The latest project report contains preliminary confounds. No listening study or checkpoint-based experiment was performed here. The recommendation's norm operates in normalized mel coordinates, where geometric distance is not perceptual distance. The student can improve on the teacher, so excessive anchoring may be counterproductive.

Conclusion. The best-supported next formulation is an endpoint acceptance operator aligned with the actual reconstruction chain. The bounded first-correction rule in Eq. (28) is a concrete candidate with an exact local geometric guarantee and zero additional neural evaluations. It is not a proven improvement in speech quality or a novel general interpolation theorem. Fixed

relaxation and no modification remain serious alternatives. Continuous cosine velocity blending should proceed only after shared-state compatibility is established and its added evaluation cost is justified. A true single-network-evaluation objective requires a suitably learned mapping rather than an algebraic combination of two frozen networks.

14. References and inspected-version notes

The publication description below refers to what was confirmed in the accessed source. An arXiv designation is not a claim that no later peer-reviewed version exists. Unresolved venue status must be checked before formal submission.

1. **R1.** Li, Y. A., Jiang, X., Tao, F., Niu, C., Xu, K., Song, J., & Mesgarani, N. (2025). *DMOSpeech 2: Reinforcement Learning for Duration Prediction in Metric-Optimized Speech Synthesis*. arXiv:2507.14988v1. Full text. Search surfaced a 2026 AAAI record, but its exact proceedings version was not resolved; equations here refer to arXiv v1 and the supplied code.
2. **R2.** Chen, Y., Niu, Z., Ma, Z., Deng, K., Wang, C., Zhao, J., Yu, K., & Chen, X. (2024). *F5-TTS: A Fairytaler that Fakes Fluent and Faithful Speech with Flow Matching*. arXiv:2410.06885. Inspected HTML v1. ACL 2025 publication was surfaced; the extraction uses the arXiv formula.
3. **R3.** Yin, T., Gharbi, M., Park, T., Zhang, R., Shechtman, E., Durand, F., & Freeman, W. T. (2024). *Improved Distribution Matching Distillation for Fast Image Synthesis*. arXiv:2405.14867. Full text.
4. **R4.** Lipman, Y., Chen, R. T. Q., Ben-Hamu, H., Nickel, M., & Le, M. (2022). *Flow Matching for Generative Modeling*. arXiv:2210.02747. Full text. Original preprint inspected; conference metadata not used for any mathematical claim.
5. **R5.** Liu, X., Gong, C., & Liu, Q. (2022). *Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow*. arXiv:2209.03003v1. Source. Title follows the inspected PDF.
6. **R6.** Lu, C., Zhou, Y., Bao, F., Chen, J., Li, C., & Zhu, J. (2022). *DPM-Solver: A Fast ODE Solver for Diffusion Probabilistic Model Sampling in Around 10 Steps*. arXiv:2206.00927. Paper; conference record.
7. **R7.** Song, Y., Dhariwal, P., Chen, M., & Sutskever, I. (2023). *Consistency Models*. Proceedings of the 40th International Conference on Machine Learning, PMLR 202, 32211–32252. Proceedings.
8. **R8.** Li, X., Bu, F., Mehrish, A., Li, Y., Han, J., Cheng, B., & Poria, S. (2024). *CM-TTS: Enhancing Real Time Text-to-Speech Synthesis Efficiency through Weighted Samplers and Consistency Models*. Findings of NAACL 2024, 3777–3794. DOI: 10.18653/v1/2024.findings-naacl.240. Proceedings.
9. **R9.** Zheng, Q., Chen, Y., Niu, Z., Ma, Z., Wang, X., Yu, K., & Chen, X. (2025). *Accelerating Flow-Matching-Based Text-to-Speech via Empirically Pruned Step Sampling*. arXiv:2505.19931. Full text.
10. **R10.** Geng, Z., Deng, M., Bai, X., Kolter, J. Z., & He, K. (2025). *Mean Flows for One-step Generative Modeling*. arXiv:2505.13447v1, technical report. Full text. A NeurIPS 2025 record was surfaced; derivations here cite the inspected technical report.
11. **R11.** Bravo, M., Cominetti, R., & Pavez-Signé, M. (2017). *Rates of convergence for inexact Krasnosel'skii–Mann iterations in Banach spaces*. arXiv:1705.09340v2. Full text.
12. **R12.** Liberzon, D., & Morse, A. S. (1999). *Basic problems in stability and design of switched systems*. IEEE Control Systems Magazine, 19(5), 59–70. Author-hosted full text.
13. **R13.** Wu, T., Li, R., Zhang, L., & Ma, K. (2026). *Diversity-Preserved Distribution Matching Distillation for Fast Visual Synthesis*. arXiv:2602.03139. Full text. Preprint source; journal/conference status not established.

14. **R14.** Wu, Y., Gao, X., Chen, T., Chen, X., Yin, Q., Tu, Z., & Lee, D. (2026). *Delta Forcing: Trust Region Steering for Interactive Autoregressive Video Generation*. arXiv:2605.14382v1. Inspected version. Later search snippets differ in description; this review uses v1 explicitly.
15. **R15.** Wang, W., Yang, Y., Deng, W., & Xu, P. (2026). *Inference-Time Alignment of Diffusion Models via Trust-Region Iterative Twisted Sequential Monte Carlo*. arXiv:2605.25123v1. Full text.
16. **R16.** dots.tts Team. (2026). *dots.tts Technical Report*. arXiv:2606.07080v1. Inspected PDF. Technical report; authorship follows the PDF masthead.
17. **R17.** Li, B., Wang, H., Xiong, H., Wu, F., Yu, J., Shi, Y., Liu, J., & Huang, R. (2026). *Rethinking Classifier-Free Guidance in On-Policy Diffusion Distillation*. arXiv:2607.24731v1. Full text. Preprint.
18. **Reporting reference.** Page, M. J., et al. (2021). *The PRISMA 2020 statement: an updated guideline for reporting systematic reviews*. BMJ, 372, n71. DOI: 10.1136/bmj.n71. Official guidance and publication links.

Appendix A. Exact search log

All searches were submitted on 19 September 2026 UTC. "Web" means the available general web search service; it is not a publisher-native database search. The domain filter for Q7–Q9 was `arxiv.org`. Quotation marks and `site:` strings below reflect the submitted query text. Queries are listed in execution order.

Query	Exact string	Purpose / result used
Q1	`"DMOSpeech 2" teacher guided sampling`	Baseline identification
Q2	`"F5-TTS" flow matching sway sampling paper`	Backbone/schedule
Q3	`"PRISMA 2020" statement systematic reviews`	Reporting guidance; primary page later opened directly
Q4	`site.arxiv.org "Flow Matching for Generative Modeling"`	Foundational field formulation
Q5	`site.arxiv.org "Improved Distribution Matching Distillation"`	Distillation mechanism
Q6	`site.arxiv.org "Consistency Models"`	Endpoint-map alternative
Q7	`Improved Distribution Matching Distillation for Fast Image Synthesis`	R3; R13 also surfaced
Q8	`Inferring switching times teacher student diffusion model inference`	Broad handover search; low precision
Q9	`Krasnoselskii Mann iteration nonexpansive operators residual`	R11
Q10	`DPM-Solver fast ODE solver diffusion probabilistic model sampling 2206.00927`	R6
Q11	`From Noise to Data with Rectified Flow 2209.03003`	Intended rectified-flow discovery; exact paper resolved by ID
Q12	`teacher student handover TTS inference cosine blending noise`	Direct claim search; low precision

Query	Exact string	Purpose / result used
Q13	`switched systems stable subsystems convex combination unstable Liberzon`	Switching foundation; low precision
Q14	`"text-to-speech" "consistency" "distillation"`	R8
Q15	`"text-to-speech" "sampling" "EPSS"`	R9
Q16	`"DMOSpeech 2" "sampling" "2026"`	Limited forward/current-status search
Q17	`"switching systems" "common Lyapunov"`	R12; related mathematical candidates
Q18	`"Mean Flows for One-step Generative Modeling"`	R10
Q19	`"CM-TTS: Enhancing Real Time Text-to-Speech Synthesis"`	R8 full publication
Q20	`"DMOSpeech 2" "aaai.org" 2026`	Publication-status check; incomplete resolution
Q21	`"PRISMA 2020" "statement" Page`	Reporting-source check
Q22	`"teacher" "student" "TTS" "handover" "noise"`	Direct causal claim search; no retained supporting study
Q23	`"DMOSpeech" "blending"`	Close prior art; no retained exact match
Q24	`"diffusion" "teacher" "student" "trust region" inference`	R14–R15; additional related candidates
Q25	`"Mean Flows" "NeurIPS" "2025"`	Publication-status check
Q26	`"DMOSpeech 2" "site:ojs.aaai.org/index.php/AAAI/article/view"`	Publication-status check; root-only result
Q27	`"F5-TTS" "ACL" 2025 Chen`	Publication-status check
Q28	`"Mean Flows for One-step Generative Modeling" "site: proceedings.neurips.cc"`	Publication-status check

Direct retrieval and reference following. Resolved arXiv identifiers were opened as full HTML or downloaded as PDFs. Some HTML endpoints failed; PDFs were successfully obtained for the technical set. CM-TTS was retrieved from ACL Anthology. The control paper was followed from Liberzon's publications page. PRISMA was retrieved from its official site after the BMJ page failed. The linked references in Section 14 specify the resulting primary sources.

Stopping rule actually used. Searching stopped after the baseline, major candidate mechanisms, mathematical foundations, and the closest discovered adaptive-guidance alternatives had accessible sources, and further status searches did not resolve additional records. This is a pragmatic scope boundary, not proof of saturation or exhaustiveness.

Appendix B. Screening and access ledger

B.1 Retained technical set

Record	Access	Decision reason	Publication status used in this review
R1	HTML and PDF	Exact baseline	Inspected arXiv v1; later venue unresolved
R2	HTML and PDF	Backbone and schedule	Inspected arXiv; ACL publication surfaced
R3	HTML and PDF	Multi-step mismatch	Preprint version inspected; venue not independently resolved
R4	HTML and PDF	Field/mixture assumptions	Preprint version inspected
R5	PDF	Reflow alternative	2022 preprint title/version inspected
R6	PDF; OpenReview record located	Numerical formulation	Conference record located; PDF equation extraction
R7	PDF; PMLR record	One-step endpoint model	Peer-reviewed ICML proceedings confirmed
R8	ACL PDF and metadata	Speech consistency model	Peer-reviewed NAACL Findings confirmed
R9	HTML and PDF	Training-free TTS efficiency	Preprint version inspected
R10	HTML and PDF	Average-velocity alternative	Tech report inspected; venue surfaced but not resolved
R11	PDF	Relaxation theorem assumptions	arXiv v2; later journal status not checked
R12	Image PDF, relevant page rendered	Switching/stability caution	IEEE magazine survey; author record confirmed
R13	HTML and PDF	Recent first-step diversity design	Preprint source; venue unverified
R14	HTML and PDF v1	Close adaptive-teacher prior art	Preprint v1; later versions not synthesized
R15	HTML and PDF v1	Constrained distribution update	Preprint v1; venue unverified
R16	PDF v1; HTML unavailable	Recent speech interval-distillation implementation	Technical report v1
R17	HTML and PDF v1	Guidance and conditioning compatibility	Preprint v1

B.2 Additional named candidates encountered

This is a record of substantively considered additional candidates, not all irrelevant web hits.

Candidate / identifier	Screening level	Decision
Laplacian Multi-scale Flow Matching, arXiv:2602.19461	Abstract	Excluded from central synthesis: different multiscale architecture; no direct handover operator
One-Step Flow Matching for Path-Dependent Physical Fields, arXiv:2606.22752	Abstract	Excluded: remote physical-field task; not needed to distinguish candidate operators
Adversarial Distribution Matching, arXiv:2507.18569	Abstract	Not extracted: training alternative; relevant to a future broader distillation review

Candidate / identifier	Screening level	Decision
DPM-Solver++, arXiv:2211.01095	Abstract	Not extracted: important numerical extension but no full-text equation used here
DPM-Solver-v3, arXiv:2310.13268	Abstract/project-page description	Not extracted: extend numerical-solver coverage in a formal SLR
Speaker Identity in Non-Verbal Vocalizations, arXiv:2606.21215	Abstract	Excluded: speaker-verification task, not synthesis handover
Path-Complete Graphs and Common Lyapunov Functions, arXiv:1612.03983	Abstract	Not extracted: R12 supplies bounded foundational context; no theorem transferred
Generic web pages, social posts, AI paper summaries	Search results	Excluded as central evidence: non-primary or irrelevant

The initial abstract/snippet screens for dots.tts and the on-policy guidance paper were subsequently upgraded to retained full-text records R16–R17. A formal expanded review should still perform complete forward-citation searches of R1/R3/R9 and broaden numerical-solver coverage. This bounded selection cannot certify novelty.

Appendix C. Formula extraction matrix

Original equations are reproduced in Section 6; the matrix makes their assumptions and computational implications comparable without treating different tasks as interchangeable.

Source / location	Original object	Shared-notation interpretation	Assumptions	Property / evidence	Computation	Applicability limit
R1 Alg. 2; P3 sampler functions	Teacher handoff plus endpoint/re-noise chain	Eqs. (2)–(4) from code	Correct checkpoint/normalization	Baseline semantics	Teacher calls + student stages	Code differs from paper shorthand
R2 Eq. 7	Sway coordinate map	$t = f_{sway}(u; s)$	Valid coefficient/grid	Sampling-placement effects	Grid arithmetic	Does not blend predictions
R3 §§4.4–4.5	Denoise/re-corrupt operator	$R_k(E)$	Stage/input-distribution compatibility	Image mismatch ablation	One generator per stage; training simulation	Not evidence of TTS handover noise
R4 Eqs. 8–9	Density-weighted field and CFM loss	$\dot{x} = v$	Correct conditional path/densities	Marginalization theorem	Training; ODE inference	Time-only blend not covered
R5 Eq. 1	Velocity regression on endpoint pairs	Straight-path target	Learned coupling/model	Reflow theory and image results	Training + ODE or distilled inference	Not frozen-network interpolation
R6 Eq. 3.4	Exact integral representation	Parameterization-specific solver	Compatible diffusion schedule; smoothness for order results	Numerical convergence framework	Multiple denoiser evaluations	Must transform prediction/time conventions
R7 Eq. 5	Boundary-constrained endpoint network	$D_S(z, t, c)$	Consistency-trained parameters	One-step image generation	One call possible after training	Boundary coefficients are not model weights
R8 Eq. 5	Speech consistency parameterization	Conditioned endpoint map	Its trained architecture	Speech experiments	Training; one/few inference calls	Not a DMOSpeech checkpoint plug-in
R9 Table 2	Pruned sway schedule	Eight nodes/seven intervals	Evaluated teacher model/settings	TTS quality–cost comparison	Reduced teacher evaluations	Student grid transfer unverified
R10 Eqs. 3,6,12	Average-velocity field	Interval map $x_b = x_a + (b - a)\bar{v}$	Learned average field	Identity; image experiments	New training; one call possible	Not arithmetic model fusion
R11 (`ikm`), Thm. 1.1	Relaxed inexact fixed-point map	Partial R_k update	Nonexpansiveness and error conditions	Residual bounds	Operator-dependent	TTS assumptions unverified
R12 Eq. 3, p. 60	Switched field family	Hypothetical Eq. (19)	Common Lyapunov conditions for cited guarantee	Stability foundation	Depends on selected subsystem	Not perceptual accuracy
R13 Eqs. 6–7	First-step teacher target/loss	Early-stage preservation via training	Shared initial noise; trained student	Image diversity study	Teacher rollout during training	Additional training required
R14 Eq. 11, v1	Feature-discrepancy sigmoid gate	Reliability-dependent weight analogy	Video feature/trajectory interpretation	Video ablations	Feature/guidance/ training costs	Does not validate mel residual magnitude
R15 Eqs. 4.5–4.6	Constrained path distribution	Distribution-space proximal update	Density ratios/support; fitted proposals	Conditional theory; text/image results	Particles + twist optimization	Not Euclidean endpoint projection
R16 Eqs. 13–14; §3.2.4	Teacher interval target and learned mean velocity	Forward-time interval update	New interval-conditioned student	TTS efficiency experiments	Additional training; one conditional call/interval	Does not demonstrate one-call full-utterance synthesis

Source / location	Original object	Shared-notation interpretation	Assumptions	Property / evidence	Computation	Applicability limit
R17 Eq. 10	Positive/direction matching loss	Branch-compatible teacher supervision	Matching CFG schema and branch meaning	Image/video conditioning studies	Training and branch evaluations	Not evidence of a handover defect in P3

Appendix D. Mathematical verification performed

Small deterministic computations checked: the default sway grid; the actual retained teacher stop under the source slicing; the resulting static call count; the raw-time cosine value; the three-stage normalized-time cosine values; and the two-coordinate bounded-correction example. The derivations in Sections 8–9 were checked algebraically. No synthetic number is presented as a checkpoint experiment.

Check	Result
First nonzero student time	0.07612046749
Raw-time cosine acceptance	0.01422892101
Default final retained teacher time	0.07612046749
Teacher RHS calls including extrapolation	5
Conditional/unconditional teacher forwards + 3 student calls	13
Illustrative cap weight	0.56568542495
Illustrative accepted RMS	2.00000000000

Adviser-facing summary

What the literature establishes. Generation formulas must match the learned prediction type and inference-state distribution. Sampling placement, training/inference mismatch, and early-stage preservation are credible research concerns. Existing work already includes relaxation, adaptive guidance, and learned one-step maps.

What remains hypothetical. We have not established that the current handover creates audible noise or that smaller first corrections improve speech. A cosine curve does not resolve those questions by itself.

What formula we should investigate. At the first student reconstruction, compute its endpoint proposal S from the cached teacher endpoint T, then accept

$$\boxed{E = T + \min\left(1, \frac{\rho}{\mathrm{RMS}(S-T)}\right)(S-T)}$$

with the zero-residual case defined as full acceptance. Compare it with a fixed weight and the original full update. Subsequent stages stay unchanged for the first experiment.

Why this is a useful candidate. It matches the inspected reconstruction algorithm, makes the allowed displacement explicit, and adds no neural evaluations. Its local bound can be proved. It remains an adaptation of known mathematics, not yet a novel or validated synthesis improvement.

What we must demonstrate. Show that the intervention improves a prespecified perceptual outcome without unacceptable losses in intelligibility, speaker identity, diversity, or runtime—and that the adaptive rule adds value over simple fixed relaxation. If it does not, the correct conclusion is to retain the simpler method or the baseline.